06 22313 Computational Tools for Modelling and Analysis

Fitting Models to Data

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Introduction

Least Squares Fitting

Linear Models

Finding the Best Fit

Weighted Linear Least Squares

Fitting Nonlinear Models

Outline

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Fitting Models to Data

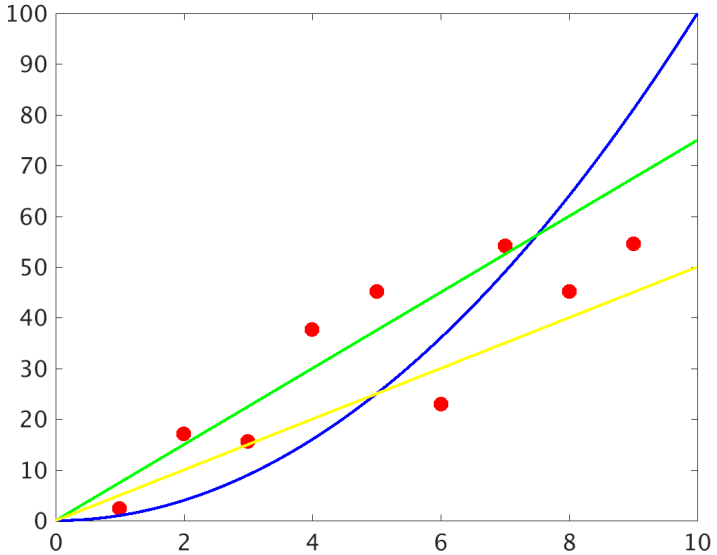
- A problem of great interest to cardiovascular scientists is to understand the dynamics of blood flow during the cardiac cycle
  - Where are the cells, how fast are they moving, are they accelerating?
- In a tracking experiment, you may be able to measure the positions of particles at set time points
- How do you calculate the velocity, acceleration, etc from the positions?
- You fit a **model** of the motion to the points, for example:

$$s = ut + \frac{1}{2}at^2,$$

where the unknowns are u, the initial velocity and a, the acceleration. These are **free parameters** that can be chosen so that the model describes the data well.

Types of Model

- How we find the optimal values of the unknowns depends on the type of model
- We will consider models of three different classes
  - Linear models:
    $y = mx + c$,
    $s = ut + \frac{1}{2}at^2$
  - Linear models with measurement error.
  - Nonlinear models:
    $y = a + b\sin(cx)$,
    $y = axe^{-a^2x}$



Least Squares Fitting

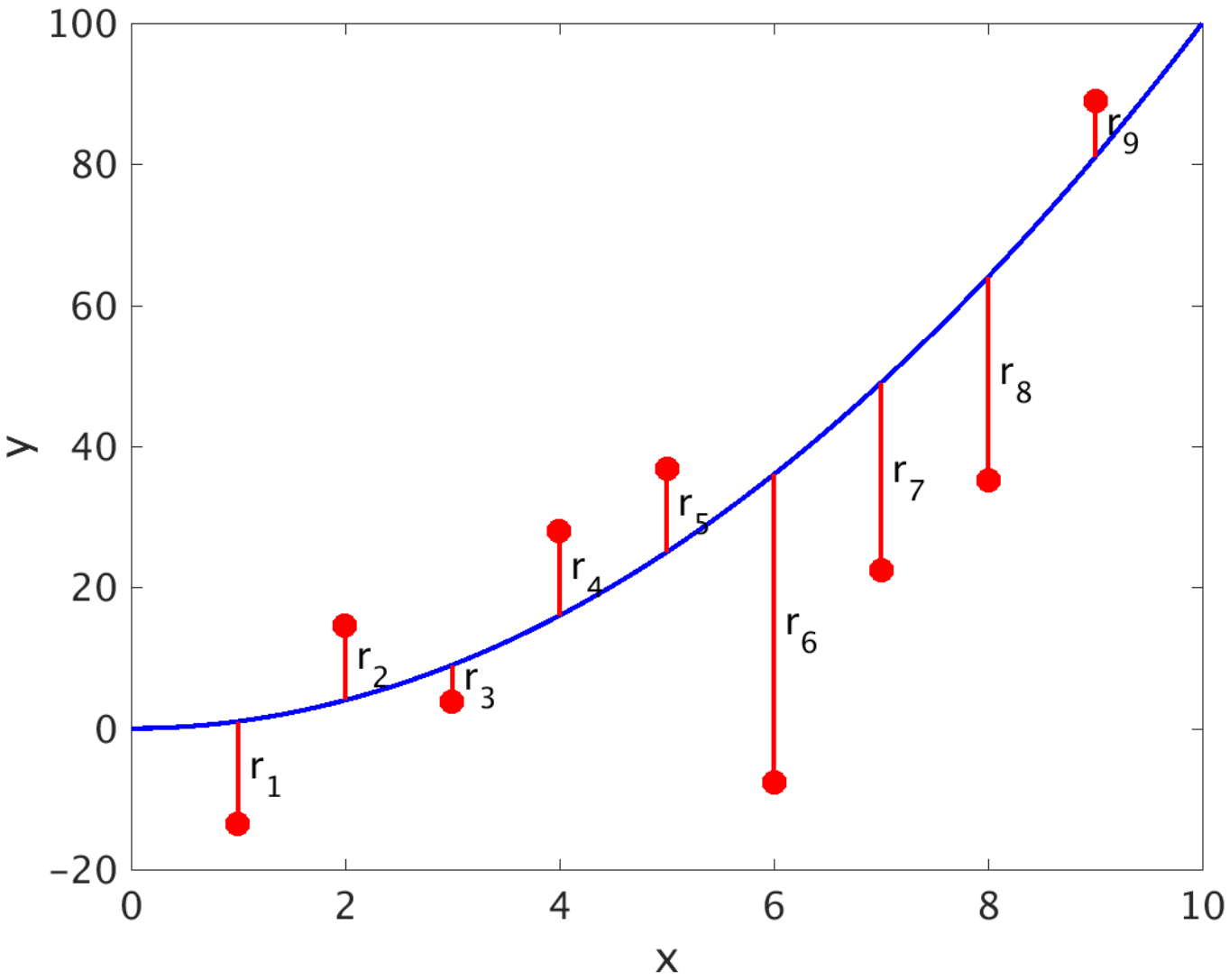
- Minimise "difference" between model and data
- Data: $\mathcal{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\}$
- Model: $y = f(\mathbf{p}, x)$
- Difference (**residual**): $r_i(\mathbf{p}) = y_i - f(\mathbf{p}, x_i)$
- In terms of which the **least squares error** is

$$e(\mathbf{p}) = \sum_{i=1}^{n} r_i^2 = \mathbf{r}^{\mathrm{T}}\mathbf{r} \tag{1}$$

- Goal: find the value of $\mathbf{p} = \mathbf{p}^*$ that minimises $e(\mathbf{p})$:

$$\mathbf{p}^* = \arg\min_{\mathbf{p}} e(\mathbf{p}) \tag{2}$$

Least Square Fitting



Linear Models

- Not just straight lines!
- Any function linear in the unknown parameters.

$$f(\mathbf{p}, x) = p_1 f_1(x) + \cdots + p_m f_m(x) = \sum_{j=1}^{m} p_j f_j(x) \quad (3)$$

- The residuals are therefore

$$r(\mathbf{p}, x_i) = y_i - \sum_{j=1}^{m} p_j f_j(x_i) \quad (4)$$

- Treating $\mathbf{p}$ as a column vector and defining matrix f $\mathbf{F}$ as $F_{ij} = f_j(x_i)$ we have

$$\mathbf{r} = \mathbf{y} - \mathbf{F}\mathbf{p} \quad (5)$$

where $\mathbf{y} = (y_1; \ldots; y_n)$.

Example: Quadratic Model

- Quadratic model $f(\mathbf{p}, x) = p_1 x^2 + p_2 x + p3$.
- Each row of $\mathbf{F}$ corresponds to a data point
- Each column corresponds to a term in the model:

$$\begin{aligned} \mathbf{r} &= \mathbf{y} - \mathbf{F}\mathbf{p} \\ \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{pmatrix} &= \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} - \begin{pmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ & \vdots & \\ x_n^2 & x_n & 1 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}. \end{aligned}$$

- Our least-squares error can therefore be written

$$e(\mathbf{p}) = \sum_i r_i^2 = \mathbf{r}^{\mathrm{T}}\mathbf{r} = (\mathbf{y} - \mathbf{F}\mathbf{p})^{\mathrm{T}} (\mathbf{y} - \mathbf{F}\mathbf{p}) \qquad (6)$$

Solving for the Best Fit

- "Best fit" $\rightarrow$ find $\mathbf{p} = \mathbf{p}^*$ that minimises $e(\mathbf{p})$.
- Differentiate $e(\mathbf{p})$ with respect to $\mathbf{p}$ and set to zero:

$$\mathbf{F}^{\mathrm{T}}\mathbf{y} - \mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{p}^* = 0$$

- Simultaneous equations: solve by direct inversion:

$$\mathbf{p}^* = \left(\mathbf{F}^{\mathrm{T}}\mathbf{F}\right)^{-1}\mathbf{F}^{\mathrm{T}}\mathbf{y}$$

- or in Matlab using Gaussian elimination through the \ operator:
- Why not solve $\mathbf{y} = \mathbf{F}\mathbf{p}$ for $\mathbf{p}$ directly?
  - Guarantees on specific solution (least squares)
  - Incorporate extra information into $e(\mathbf{p})$

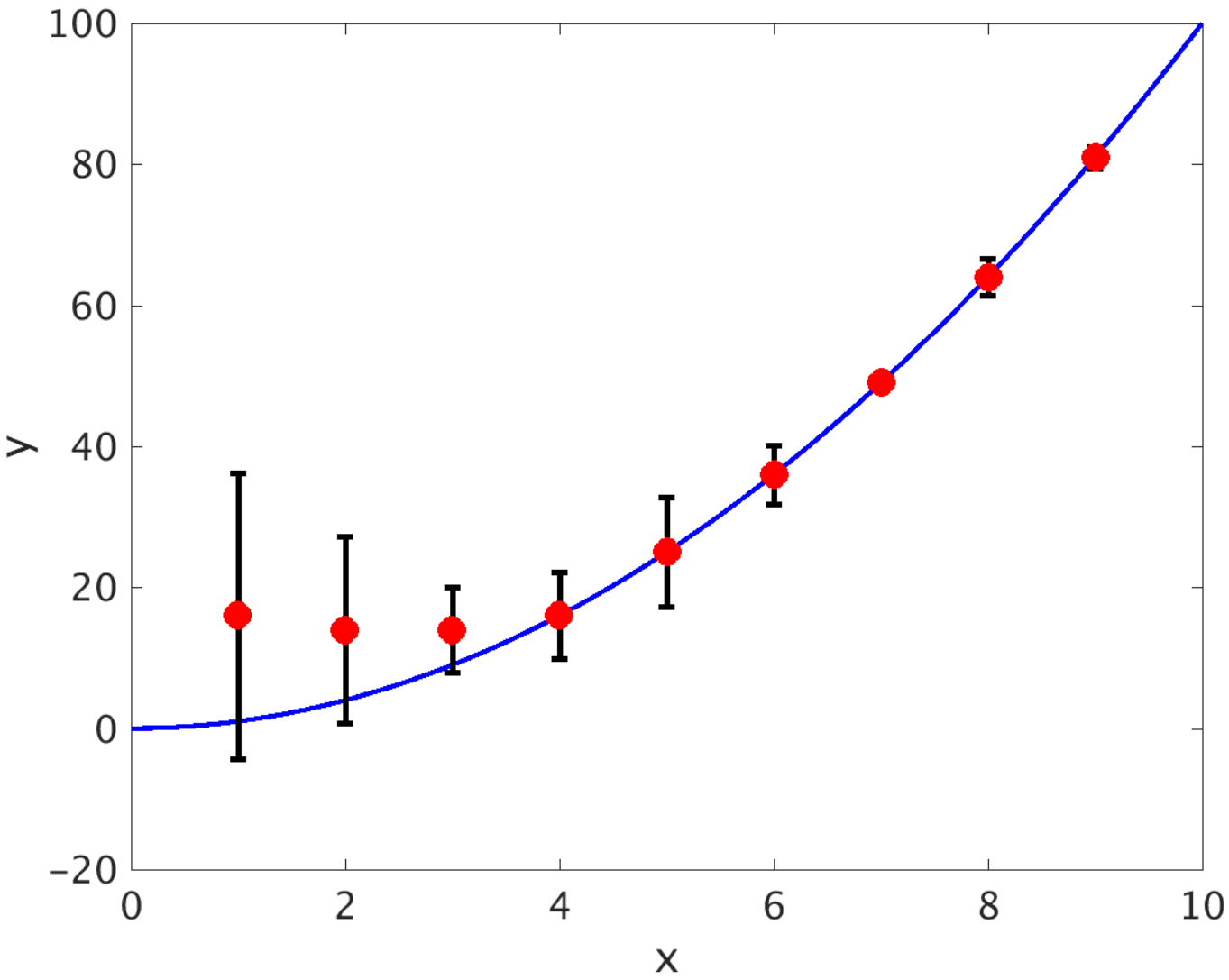
Weighted Least Squares

- Experimental data $\rightarrow$ multiple repeats. How do we deal with this?
- Fit models to the "average" but does the spread (variance) matter?
- Given less influence to "noisy/uncertain" data by modifying $e(\mathbf{p})$

$$e(\mathbf{p}) = \sum_{i=1}^{n} w_i r_i^2$$

- Set $w_i = \frac{1}{\mathrm{var}(y_i)}$
- "Certain" points influence the fit more strongly than "uncertain" ones.

Weighted Least Squares



Weighted Least Squares

- Covenient to define matrix

$$\mathbf{\Sigma} = \mathrm{diag}\left(\mathrm{var}(y_1), \mathrm{var}(y_1), \ldots, \mathrm{var}(y_n)\right)$$

- Special case of the **covariance matrix**
- The error becomes

$$e(\mathbf{p}) = \mathbf{r}^{\mathrm{T}}\mathbf{\Sigma}^{-1} = (\mathbf{y} - \mathbf{F}\mathbf{p})^{\mathrm{T}}\,\mathbf{\Sigma}^{-1}\,(\mathbf{y} - \mathbf{F}\mathbf{p})$$

- Note that $\mathbf{\Sigma}^{-1}$ can be done by inverting the diagonal elements
- Differentiate and set to zero

$$\mathbf{F}^{\mathrm{T}}\mathbf{\Sigma}^{-1}\mathbf{y} - \mathbf{F}^{\mathrm{T}}\mathbf{\Sigma}^{-1}\mathbf{F}\mathbf{p}^{*} = 0, \tag{7}$$

- Solve as before

$$\mathbf{p}^{*} = (\mathbf{F}^{\mathrm{T}}\mathbf{\Sigma}^{-1}\mathbf{F})^{-1}\mathbf{F}^{\mathrm{T}}\mathbf{\Sigma}^{-1}\mathbf{y} \tag{8}$$

Fitting Nonlinear Models

- What about non-linear models like $y(x) = a \sin bx$?
- No matrix representation (non-linear!)
- Cannot minimise least square error directly.
- Instead, focus on **reducing** the error iteratively
- We will turn non-linear problems into locally linear ones.

Linearising Non-linear Problems

Add figure in here that shows how to Taylor-expand a 1d function

- Taylor Expansion

$$y(x + \Delta x) = y(x) + \frac{\mathrm{d}y}{\mathrm{d}x}\Delta x$$

Linearising Non-linear Problems

- Taylor expansion of our fitting function

$$\begin{aligned} f(x_i, \mathbf{p} + \mathbf{\Delta p}) &\approx f(x_i, \mathbf{p}) + \sum_{j=1}^{m} \frac{\partial f(x_i)}{\partial p_j} \Delta p_j \\ &= f(x_i, \mathbf{p}) + \sum_{j=1}^{m} J_{ij} \delta p_j \end{aligned}$$

- Writing $f_i(\mathbf{p}) = f(x_i, \mathbf{p})$ give $\mathbf{f}(\mathbf{p} + \mathbf{\Delta p}) = \mathbf{f}(\mathbf{p}) + \mathbf{J}\mathbf{\Delta p}$
- The residual is $\mathbf{r}(\mathbf{p} + \mathbf{\Delta p}) = \mathbf{y} - \mathbf{f}(\mathbf{p}) - \mathbf{J}\mathbf{\Delta p}$
- And the error becomes

$$e(\mathbf{p} + \mathbf{\Delta p}) = (\mathbf{y} - \mathbf{f}(\mathbf{p}) - \mathbf{J}\mathbf{p})^{\mathrm{T}} (\mathbf{y} - \mathbf{f}(\mathbf{p}) - \mathbf{J}\mathbf{p}) .$$

- Minimise this $\rightarrow$ change in $\mathbf{p}$ that maximally reduces the error
- Differentiate and set to zero:

$$\left(\mathbf{J}^{\mathrm{T}}\mathbf{J}\right) \mathbf{\Delta p}^{*} = \mathbf{J}^{\mathrm{T}} \left(\mathbf{y} - \mathbf{f}(\mathbf{p})\right)$$

- Solve, repeat until finished!

Stabilising the Optimisation

- Common modification:

$$\left(\mathbf{J}^{\mathrm{T}}\mathbf{J} + \lambda\,\mathrm{diag}(\mathbf{J}^{\mathrm{T}}\mathbf{J})\right)\boldsymbol{\Delta}\mathbf{p}^{*} = \mathbf{J}^{\mathrm{T}}\left(\mathbf{y} - \mathbf{f}(\mathbf{p})\right)$$

- Without this modification, the parameter vector will change quickly along directions with steep gradients
- But slowly along shallow gradients.
- Modification increases movement along shallow gradients and thus faster convergence.
- "Damping factor" λ controls the extent of this.
- Modified iteratively to give the best results.
- The full algorithm is named after its inventors: **Levenberg-Marquadt**

The Levenberg-Marquadt Algorithm

```
Data: Data x, y; Initial parameter guess p_0; Function
      f(x, p); Jacobian Jac(x, p); initial values of λ, ν.
Result: p, e(p), parameters that minimmise the least
        squares error, the error.
% Evaluate the error at the initial parameter
  value
p ← p_0;
e ← (y − f(x, p)^T (y − f(x, p);
% Iterate until convergence
Converged ← False;
while Converged == False do
    % Calculate the Jacobian
    J ← Jac(x, p);
    % Calculate update for different values of λ
      until we improve the error
    LambdaSet ← false;
    while LambdaSet==false do
        % Calculate Δp for λ = λ_0 and λ = λ_0/ν
        Δ_1 = (J^T J + λ diag(J^T J)) \ J^T (y − f(p))%;
        Δ_2 = (J^T J + (λ/ν) diag(J^T J)) \ J^T (y − f(p));
        % Calculate the error at the new
          parameter values
        e_1 ← (y − f(x, p + Δ_1)^T (y − f(x, p);
        e_2 ← (y − f(x, p + Δ_2)^T (y − f(x, p);
        if e_1 < e then
            % Set new values of p and e, storing
              old value as e'
            p ← p + Δ_1;
            e' ← e;
            e ← e_1;
            % Current λ is OK
            LambdaSet ← true;
        else if e_2 < e then
            % Set new values of p and e, storing
              old value as e'
            p ← p + Δ_2;
            e' ← e;
            e ← e_2;
            λ ← λ/ν;
            % Current λ is OK
            LambdaSet ← true;
        else
            λ ← λ/ × nu;
    end
    if e'/e < TerminationCriterion then
        Converged ← true
    end
end
```